

Xuewen Luo

PhD Student, Transportation Engineering
University of Utah

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EDUCATION

University of Utah

Ph.D. in Transportation Engineering

08/2025 – Present

- Research Focus: Autonomous Driving, Human–Computer Interaction (HCI), Intelligent Transportation Systems (ITS), Vehicle-to-Everything (V2X) Cooperative Perception, and Vision–Language Models (VLMs)
- Advisor: Prof. Chenxi (Dylan) Liu
- Current Work: V2X-UniPool and Human-in-the-Autonomous-Driving-Loop (HADL)

Monash University

Master of Artificial Intelligence

07/2023 – 07/2025

- Reward: High Distinction
- Dissertation: Computer Architecture Networks, Machine Learning, Deep Learning, Introduction to Database, IT Research Methods, Modeling Discrete Optimization Problems

Yunnan University

Bachelor of Logistic Management

09/2018 – 06/2022

- Dissertation: Advanced Mathematics, Fundamentals of Programming Design, Application of Lean Transformation Information Technology, Fundamentals of Computer, Logistics System Simulation

RESEARCH EXPERIENCE

Research on End-to-End Autonomous Driving

Monash University

Research Intern (Supervisor: Junnyong Loo)

10/2023 – 06/2025

- Conducted literature reviews on autonomous driving, deep learning, and computer vision.
- Reproduced BEVFormer and BEVFusion models, performing simulations on MATLAB and PyTorch.
- Published work: *Energy-efficient Hybrid Model Predictive Planner for Autonomous Driving* (SMC 2024).

Research on Smart Transportation Systems for Intersection Safety

University of Utah

Research Intern (Supervisor: Chenxi Liu)

07/2024 – 10/2024

- Trained YOLO models for object detection and implemented object tracking using StrongSORT.
- Developed predictive models for intersection collision scenarios using real-time traffic data.

Research on Driving Behavior Evaluation

Tsinghua University

Research Intern (Supervisor: Jiangtao Gong)

06/2024 – 10/2024

- Processed predictive driving behavior data and created knowledge graphs for analysis.
- Utilized large language model RAG-KG to evaluate driving behavior comprehensively.
- Paper: Accepted By ICRA 2025.

REFEREED PUBLICATIONS

Peer-Reviewed Journal and Conference Publications

1. **X. Luo**, F. Yang, F. Ding, X. Gao, B. Yu, Z. Li, Z. Tu, Y. Zhou, C. Liu. “V2X-UniPool: Unifying Multimodal Perception and Knowledge Reasoning for Autonomous Driving.” *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pp. 747–756, 2026.
2. F. Yang, B. Yu, Y. Zhou, **X. Luo**, Z. Tu, C. Liu. “Edge-Based Multimodal Sensor Data Fusion with Vision-Language-Action Models for Real-Time Autonomous Vehicle Accident Avoidance.” *Engineering Applications of Artificial Intelligence*, vol. 180, Art. 115186, 2026.
3. **X. Luo**, F. Ding, Y. Song, X. Zhang, J. Loo. “PKRD-CoT: A Unified Chain-of-Thought Prompting for Multi-Modal Large Language Models in Autonomous Driving.” *International Conference on Neural Information Processing (ICONIP)*, pp. 62–76, 2024.
4. **X. Luo**, F. Ding, F. Yang, Y. Zhou, J. Loo, H. H. Tew, C. Liu. “SenseRAG: Constructing Environmental Knowledge Bases with Proactive Querying for LLM-Based Autonomous Driving.” *IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW)*, pp. 989–996, 2025.
5. **X. Luo**, F. Ding, R. Chen, R. Panda, J. Loo, S. Zhang. “What’s Happening—A Human-Centered Multimodal Interpreter Explaining the Actions of Autonomous Vehicles.” *IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW)*, pp. 1073–1080, 2025.
6. S. You, **X. Luo**, X. Liang, J. Yu, C. Zheng, J. Gong. “A Comprehensive LLM-Powered Framework for Driving Intelligence Evaluation.” *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.
7. F. Ding, **X. Luo**, G. Li, H. H. Tew, J. Y. Loo, W. T. Chor, A. S. M. Bakibillah, Z. Zhao, Z. Tao. “Energy-Efficient Hybrid Model Predictive Trajectory Planning for Autonomous Electric Vehicles.” *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, pp. 1263–1269, 2024.
8. J. Q. Loh, **X. Luo**, F. Ding, H. H. Tew, J. Y. Loo, Z. Y. Ding, S. Susilawati, C. P. Tan. “Cross-Domain Transfer Learning Using Attention Latent Features for Multi-Agent Trajectory Prediction.” *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, 2024.
9. H. H. Tew, G. Li, F. Ding, **X. Luo**, J. Y. Loo, C. M. Ting, Z. Y. Ding, C. P. Tan. “KANS: Knowledge Discovery Graph Attention Network for Soft Sensing in Multivariate Industrial Processes.” *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, pp. 4377–4383, 2024.
10. G. Li, C. H. Lim, Q. Ma, X. Tang, H. H. Tew, F. Ding, **X. Luo**. “FedBChain: A Blockchain-Enabled Federated Learning Framework for Improving DeepConvLSTM with Comparative Strategy Insights.” *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, pp. 2607–2612, 2024.

Preprints

1. **X. Luo**, F. Ding, R. Chen, Y. Cao, X. Liu, B. Yu, F. Yang, C. Liu. “Who Responds When the Driver Is Gone? A Framework for Human Intent Understanding.” *arXiv preprint arXiv:2607.04670*, 2026.
2. F. Yang, B. Yu, **X. Luo**, C. Liu, C. Liu. “CIVIC: End-to-End Sequence Compactness for Efficient Vision-Language Models.” *arXiv preprint arXiv:2605.28115*, 2026.
3. F. Ding, **X. Luo**, F. Yang, B. Yu, H. H. Tew, G. Krishnasamy, J. Y. Loo. “KnowDiffuser: A Knowledge-Guided Diffusion Planner with LLM Reasoning.” *arXiv preprint arXiv:2603.10441*, 2026.
4. B. Yu, F. Yang, Y. Liu, C. Wang, **X. Luo**, T. Li, R. Ke, X. Zhou, C. Liu. “Locatability-Guided Adaptive Reasoning for Image Geo-Localization with Vision-Language Models.” *arXiv preprint arXiv:2603.13628*, 2026.
5. **X. Luo**, F. Yang, F. Ding, H. F. Yang, Z. Wang, C. Liu. “Talk then Cooperate: Language as a Unified and Efficient Intermediate Representation in Cooperative Autonomous Driving.” Under review at *Transportation Research Part C: Emerging Technologies*; SSRN preprint 6517905, 2026.

6. F. Ding, **X. Luo**, H. H. Tew, R. Reddy, X. Wang, J. Y. Loo. "Drive As You Like: Strategy-Level Motion Planning Based on a Multi-Head Diffusion Model." *arXiv preprint arXiv:2508.16947*, 2025.
7. F. Ding, H. H. Tew, J. Y. Loo, S. Susilawati, L. T. Liu, F. Y. Leong, **X. Luo**, K. K. Chin, J. J. Gan. "GSMT: Graph Fusion and Spatiotemporal Task Correction for Multi-Bus Trajectory Prediction." *arXiv preprint arXiv:2508.09227*, 2025.
8. X. Gao, Y. Wu, **X. Luo**, K. Wu, X. Chen, Y. Wang, C. Liu, Y. Zhou, Z. Tu. "AirV2X: Unified Air-Ground Vehicle-to-Everything Collaboration." *arXiv preprint arXiv:2506.19283*, 2025.

WORK & INTERSHIP EXPERIENCE

Research Intern

Glydways

In-Cabin Passenger Perception and Context Understanding

08/2026 – 12/2026

- Developing in-cabin perception methods to detect and understand passenger presence, behavior, and contextual cues in autonomous vehicles.
- Investigating context-aware passenger understanding to support safer, more responsive, and human-centered mobility experiences.
- Exploring multimodal approaches for interpreting passenger needs and interactions in dynamic in-vehicle environments.

LLM Algorithm Intern

Li Auto (Beijing) China

Intern

12/2024 – 03/2025

- Synthesized Chain-of-Thought (CoT) text data to enhance large model reasoning capabilities.
- Investigated effective strategies for Supervised Fine-tuning (SFT), such as multi-agent model integration.
- Conducted fine-tuning experiments, analyzed results, and identified performance improvements.

PROFESSIONAL SERVICES

Transportation Research Board (TRB)

Friend Member, Connected and Automated Vehicle Systems Committee

Friend Member, Integrated Transportation Safety Management Committee

American Society of Civil Engineers (ASCE)

Corresponding Member, Data Sensing & Analytics Committee

Corresponding Member, Transportation Safety Committee

REFeree FOR JOURNALS AND CONFERENCES

Conferences

Winter Conference on Applications of Computer Vision (WACV) 2025

COTA International Conference of Transportation Professionals (CICTP) 2025

International Conference on Intelligent Computing (ICIC) 2025

Transportation Research Board (TRB) Annual Meeting 2026

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026

Conference on Neural Information Processing Systems (NeurIPS) 2026

Transportation Research Board (TRB) Annual Meeting 2027 — Paper Review Coordinator (PRC)

Journal

Frontiers of Information Technology & Electronic Engineering